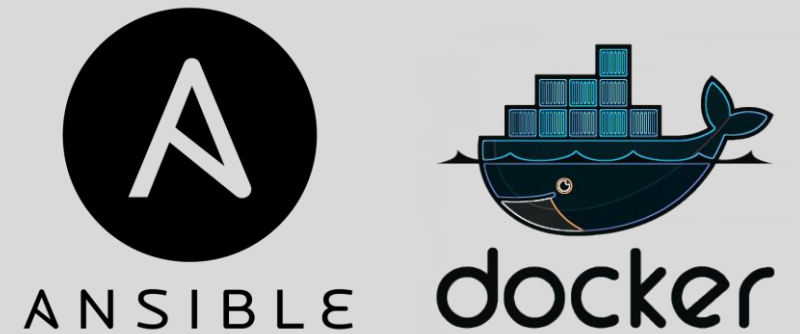
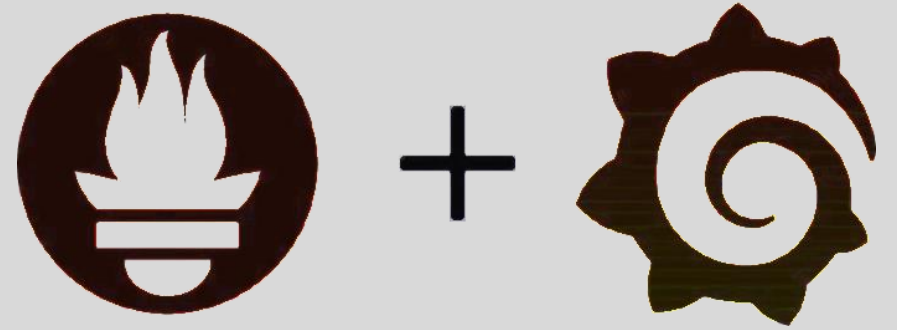


Setting up Infrastructure Monitoring Platform using Terraform, Ansible, Docker & GitLab CI/CD

Infrastructure as Code | AWS | DevOps | Monitoring | Automation

Presented by Sazad Ahemad

gitlab.com/connect2sazad/prometheus-grafana-infra-setup

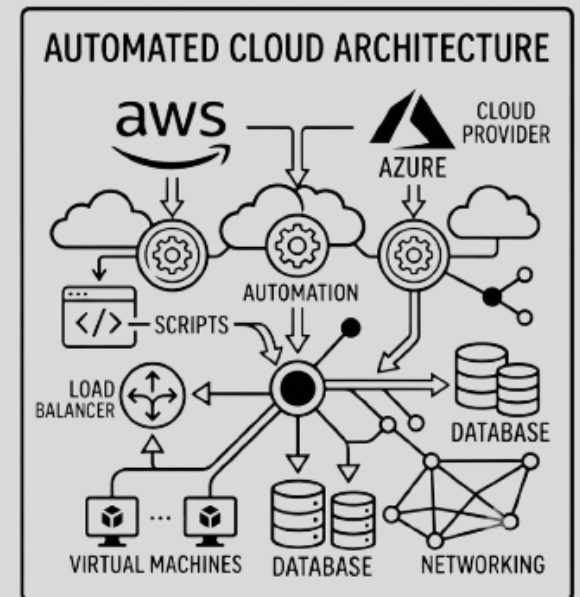
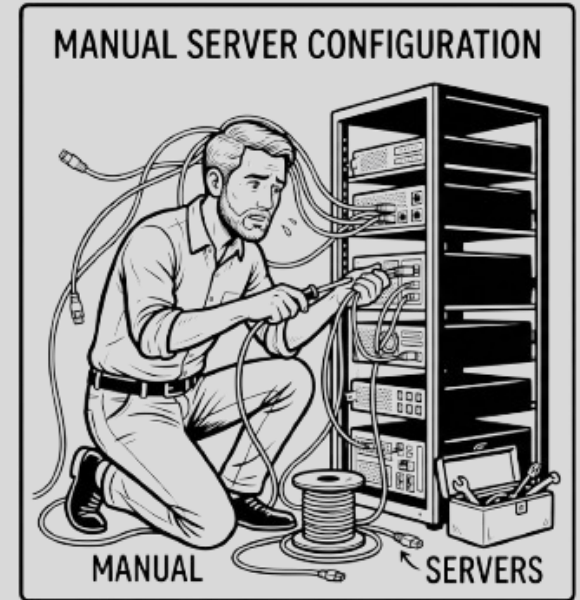


Problem Statement

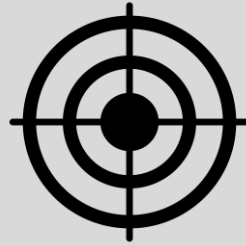


Traditional Infrastructure deployment has problems:

- Manual Deployment
- Human Errors
- No Monitoring
- Difficult Scaling
- No Automation
- Difficult Resource Management
- Difficult Tracking Resources in case of Large Infrastructure Plans



Project Objectives



Project is made to satisfy the below goals:

- CI/Cd to setup the whole infra
- Implement IaC to provision infra in a cloud provider (AWS in this case)
- Configuring Monitoring Server
- Setting up Monitoring tools
- Alert Management
- Dashboard Setup

GitLab CI/CD

Terraform

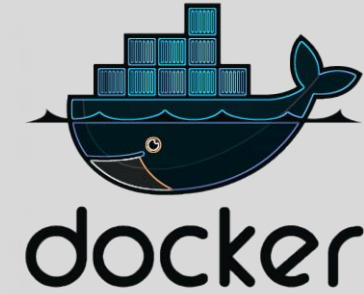
AWS

Ansible

Docker

Monitoring Server

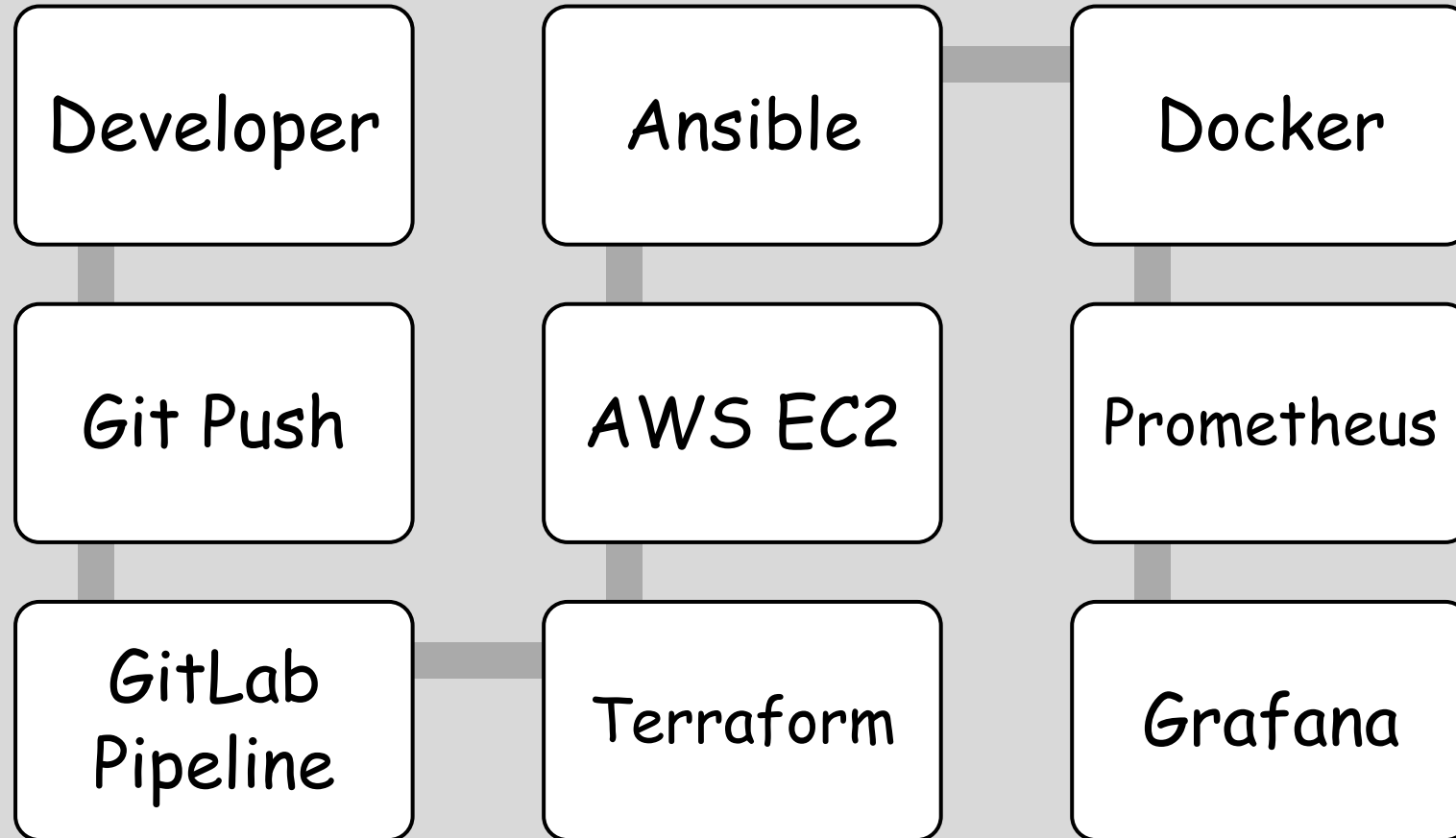
Technology Stack



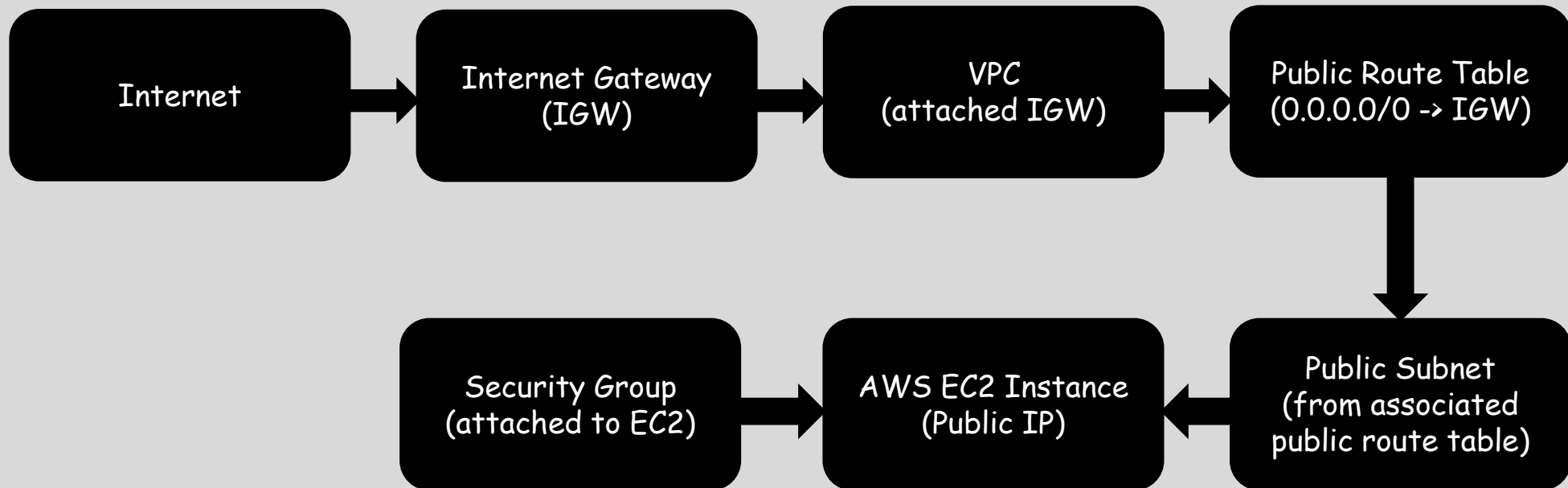
+



High Level Architecture



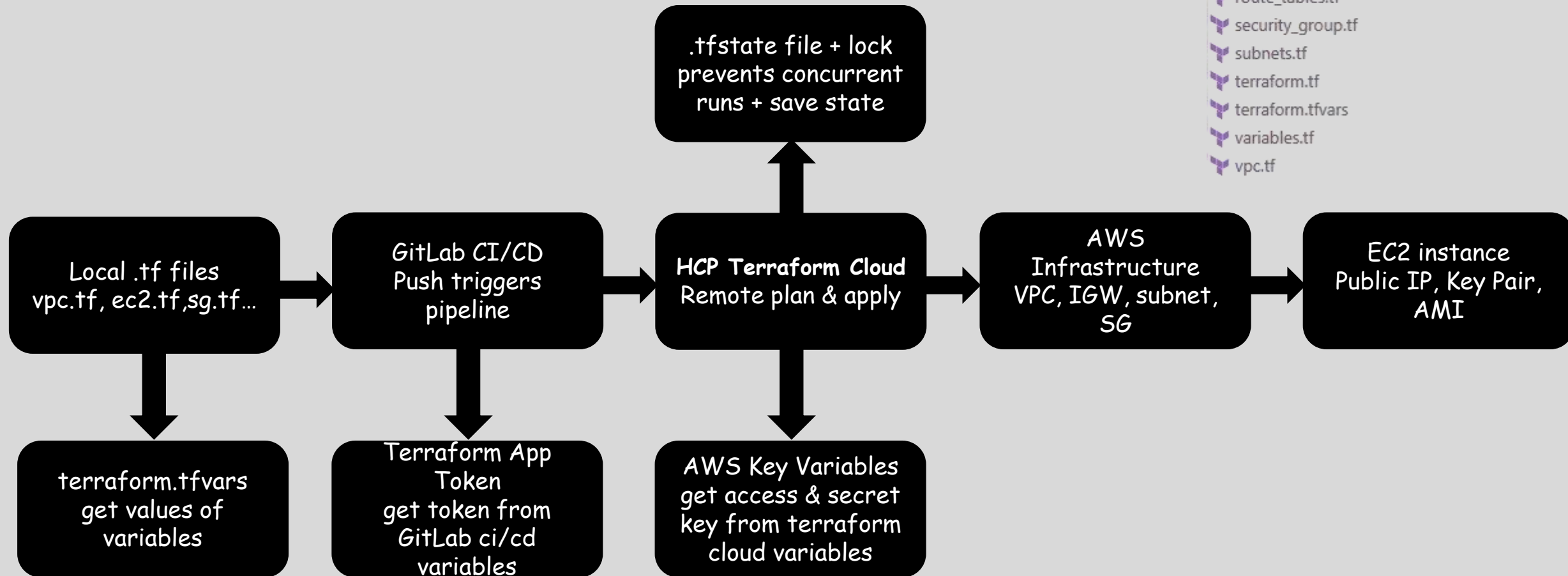
AWS Infrastructure



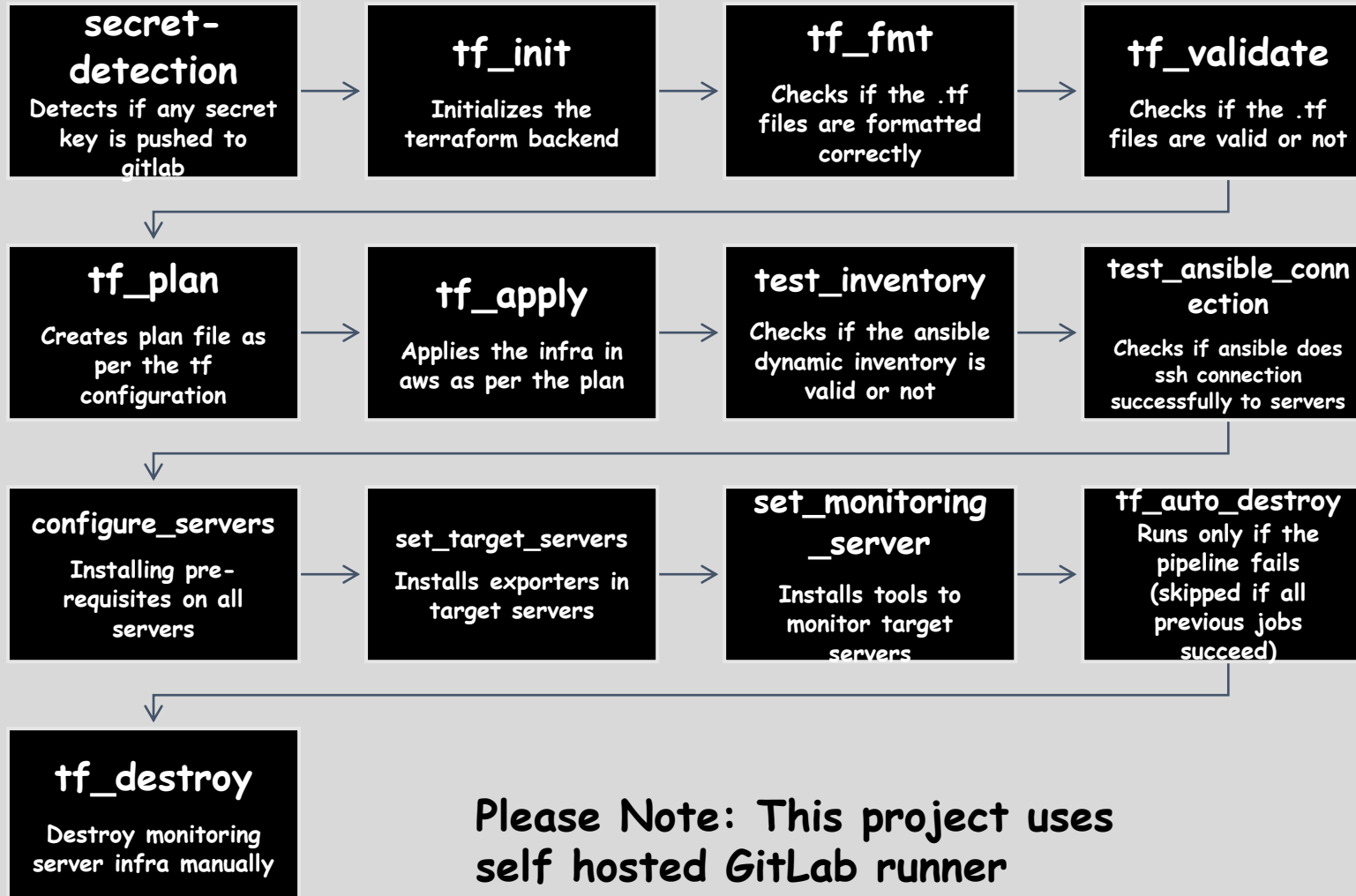
Terraform Infrastructure



- terraform
 - ami.tf
 - ec2.tf
 - internet_gateway.tf
 - key_pair.tf
 - outputs.tf
 - provider.tf
 - route_tables.tf
 - security_group.tf
 - subnets.tf
 - terraform.tf
 - terraform.tfvars
 - variables.tf
 - vpc.tf

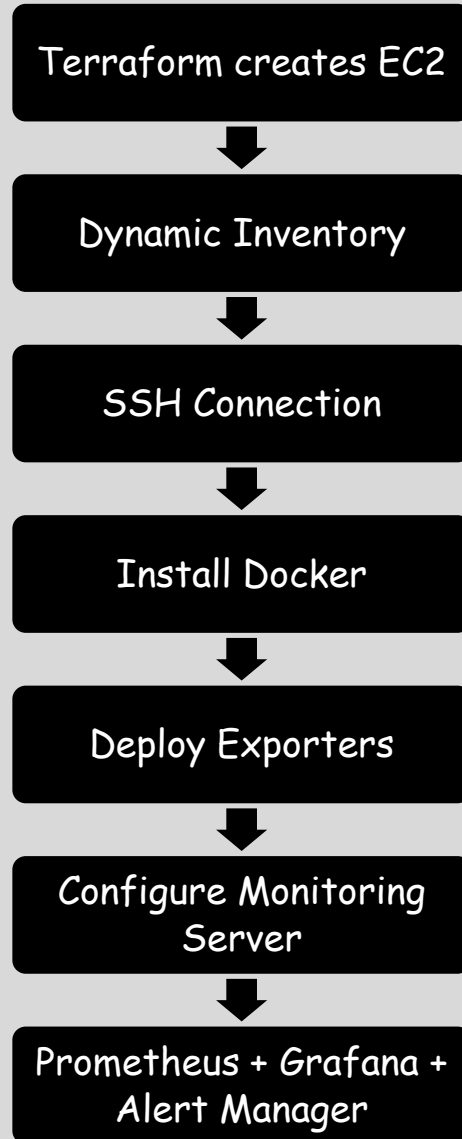


GitLab CI/CD Flow

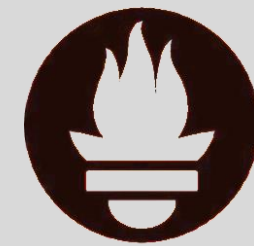
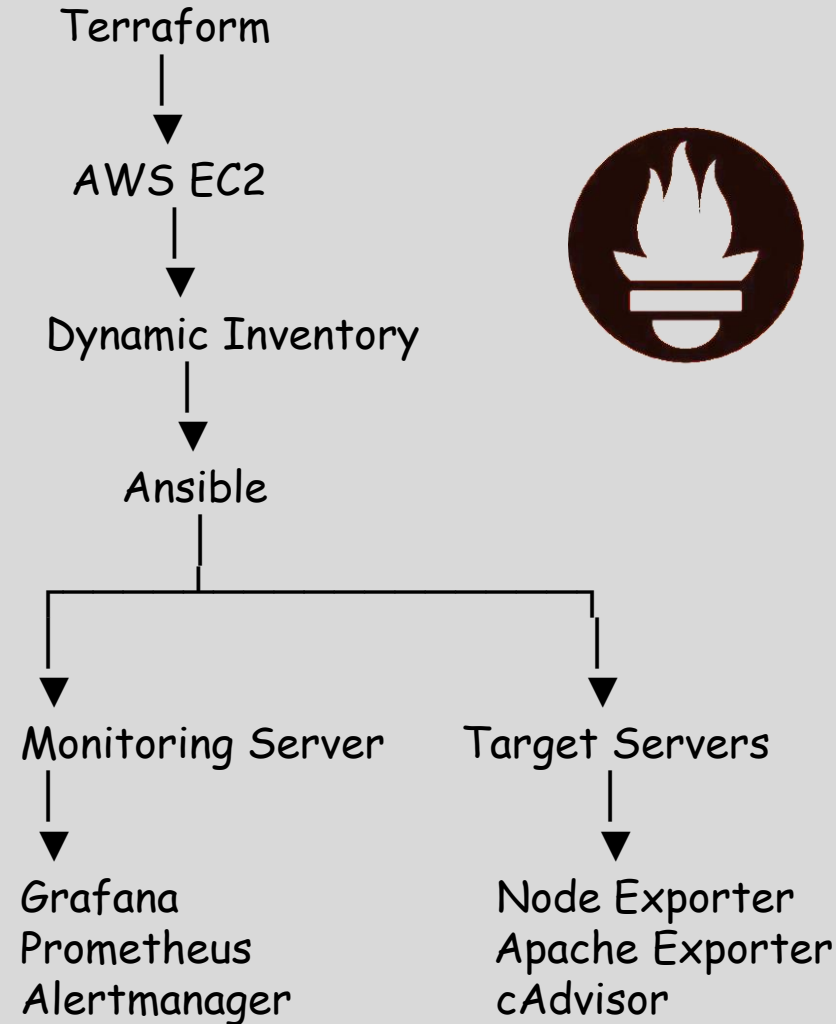


Manual	#1553765987: tf_destroy	tf_destroy
	✓ master → 75612c32	
	aws-runner docker self-hosted allowed to fail manual	
Skipped	#1553765987: tf_auto_destroy	tf_auto_destroy
	✓ master → 75612c32	
	aws-runner docker self-hosted	
Passed	#1553765986: set_monitoring_server	set_monitoring_s...
00:02:46	✓ master → 75612c32	
1 week ago	aws-runner docker self-hosted	
Passed	#1553765986: set_target_servers	set_target_servers
00:01:50	✓ master → 75612c32	
1 week ago	aws-runner docker self-hosted	
Passed	#1553765986: configure_servers	configure_servers
00:02:46	✓ master → 75612c32	
1 week ago	aws-runner docker self-hosted	
Passed	#1553765986: test_ansible_connection	test_ansible_conn...
00:01:32	✓ master → 75612c32	
1 week ago	aws-runner docker self-hosted	
Passed	#1553765986: test_inventory	test_inventory
00:01:34	✓ master → 75612c32	
1 week ago	aws-runner docker self-hosted	
Passed	#1553765986: tf_apply	tf_apply
00:01:36	✓ master → 75612c32	
1 week ago	aws-runner docker self-hosted	
Passed	#1553765986: tf_plan	tf_plan
00:01:21	✓ master → 75612c32	
1 week ago	aws-runner docker self-hosted	
Passed	#1553765986: tf_validate	tf_validate
00:02:46	✓ master → 75612c32	
1 week ago	aws-runner docker self-hosted	
Passed	#1553765986: tf_fmt	tf_fmt
00:00:31	✓ master → 75612c32	
1 week ago	aws-runner docker self-hosted	
Passed	#1553765986: tf_init	tf_init
00:00:50	✓ master → 75612c32	
1 week ago	aws-runner docker self-hosted	
Passed	#1553765985: secret_detection	secret-detection
00:00:24	✓ master → 75612c32	
1 week ago	aws-runner docker self-hosted	

Ansible Configuration & Automation



```
roles/  
docker  
|— install docker  
|— docker compose  
  
node_exporter  
|— docker compose  
|— start exporter  
  
apache_exporter  
|— docker compose  
  
cadvisor  
|— docker compose  
  
prometheus  
|— prometheus.yml  
|— alert rules  
  
grafana  
|— datasource  
|— dashboards  
  
alertmanager  
|— alertmanager.yml
```

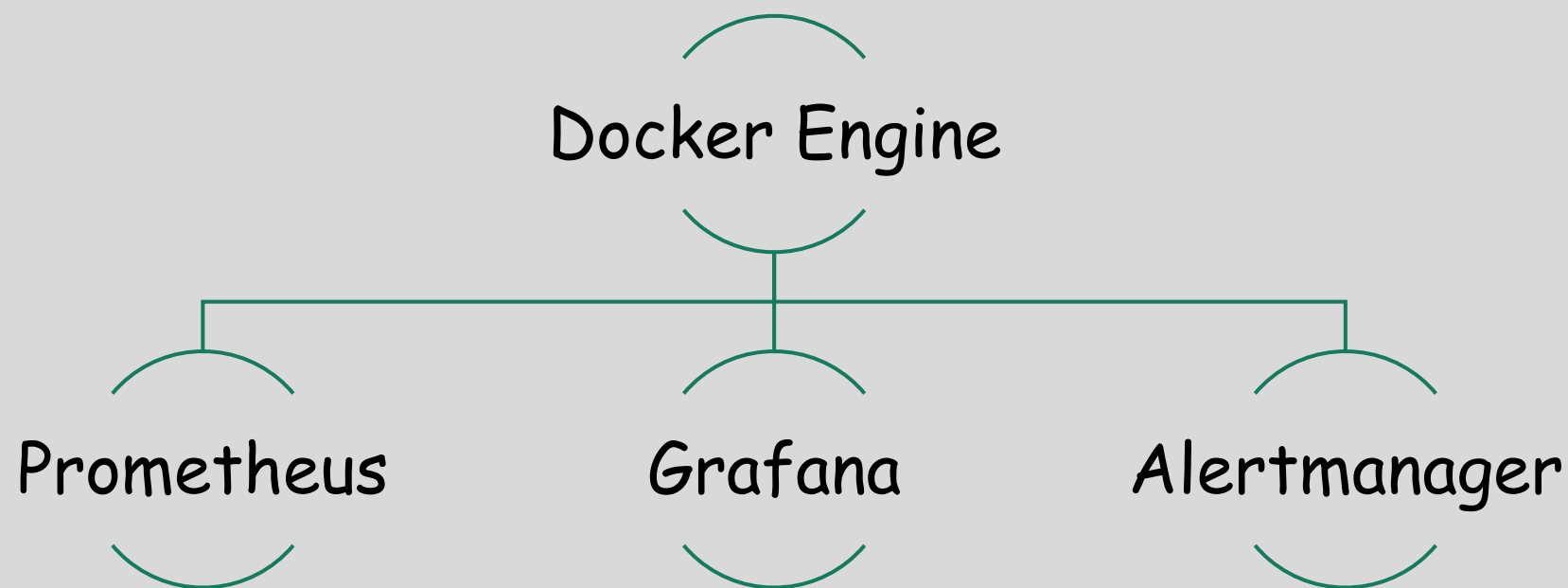
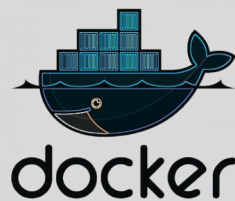


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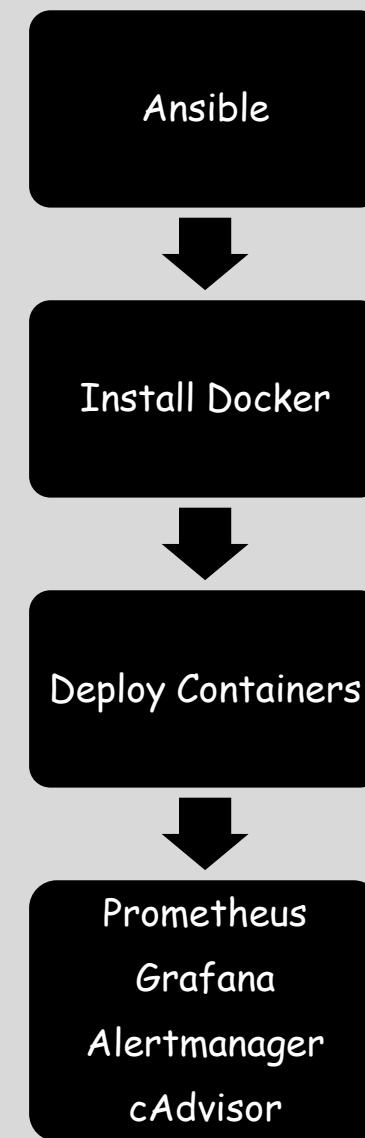


Alertmanager

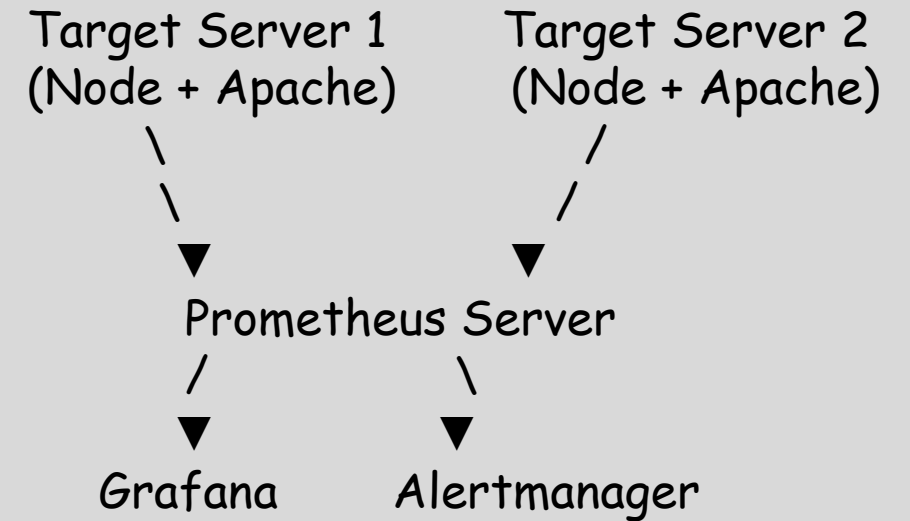
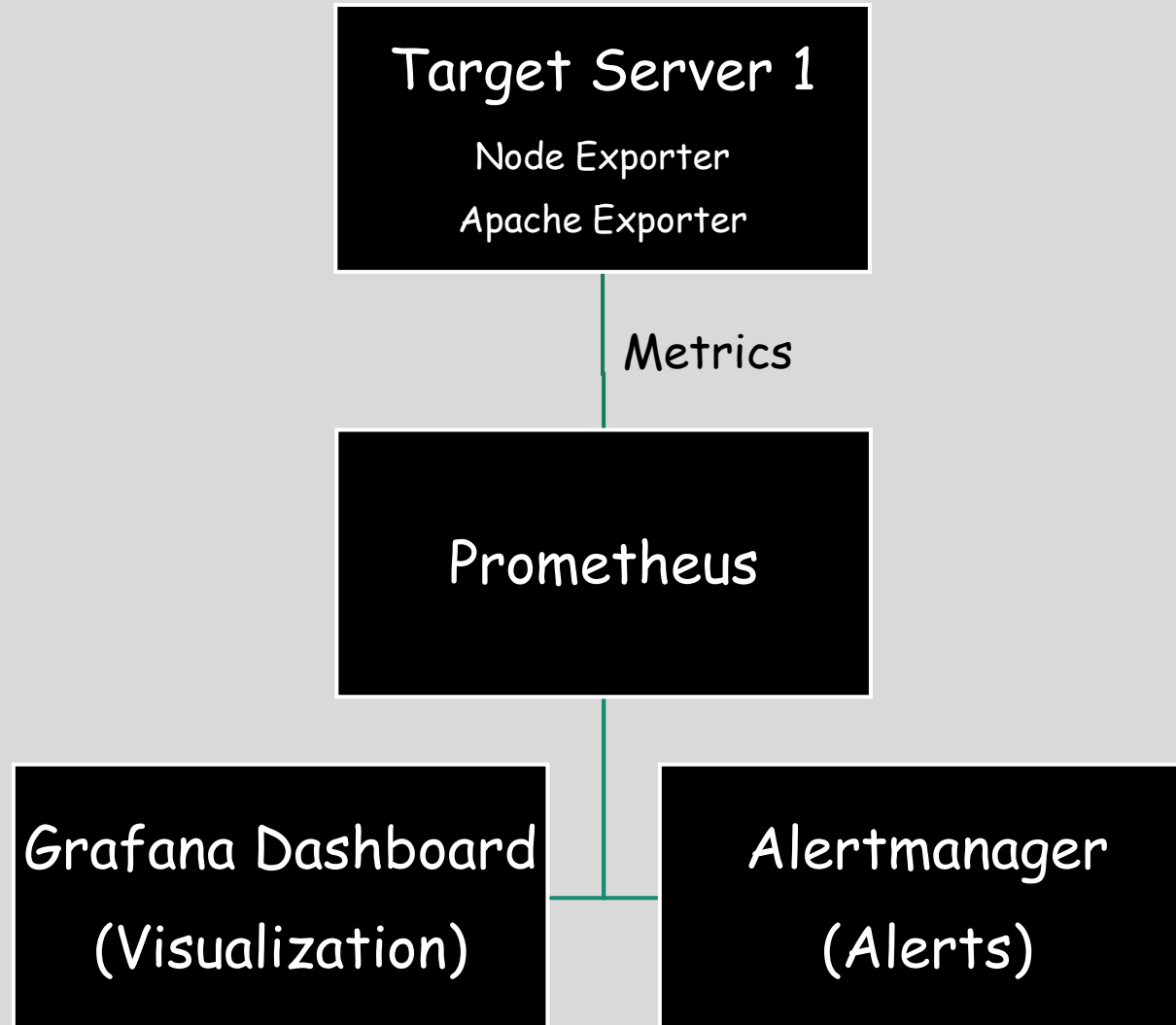
Docker Deployment



- ✓ Lightweight containerized deployment
- ✓ Simplified application management
- ✓ Consistent runtime environment
- ✓ Easy scaling and maintenance



Monitoring Architecture



Monitoring Flow

- Exporters collect system metrics
- Prometheus scrapes metrics
- Grafana visualizes dashboards
- Alertmanager handles alerts

Prometheus



Responsibilities

- ✓ Collect Metrics
- ✓ Store Time-series Data
- ✓ Evaluate Alert Rules
- ✓ Serve Data to Grafana

Node Exporter
Apache Exporter
cAdvisor



Prometheus



Stores Metrics

The screenshot displays the Prometheus web interface with the 'Status > Target health' tab selected. It shows a list of scrape pools and their individual targets. Each target row includes the endpoint, labels (instance and job), the last scrape time, the scrape duration, and the current state (UP or DOWN).

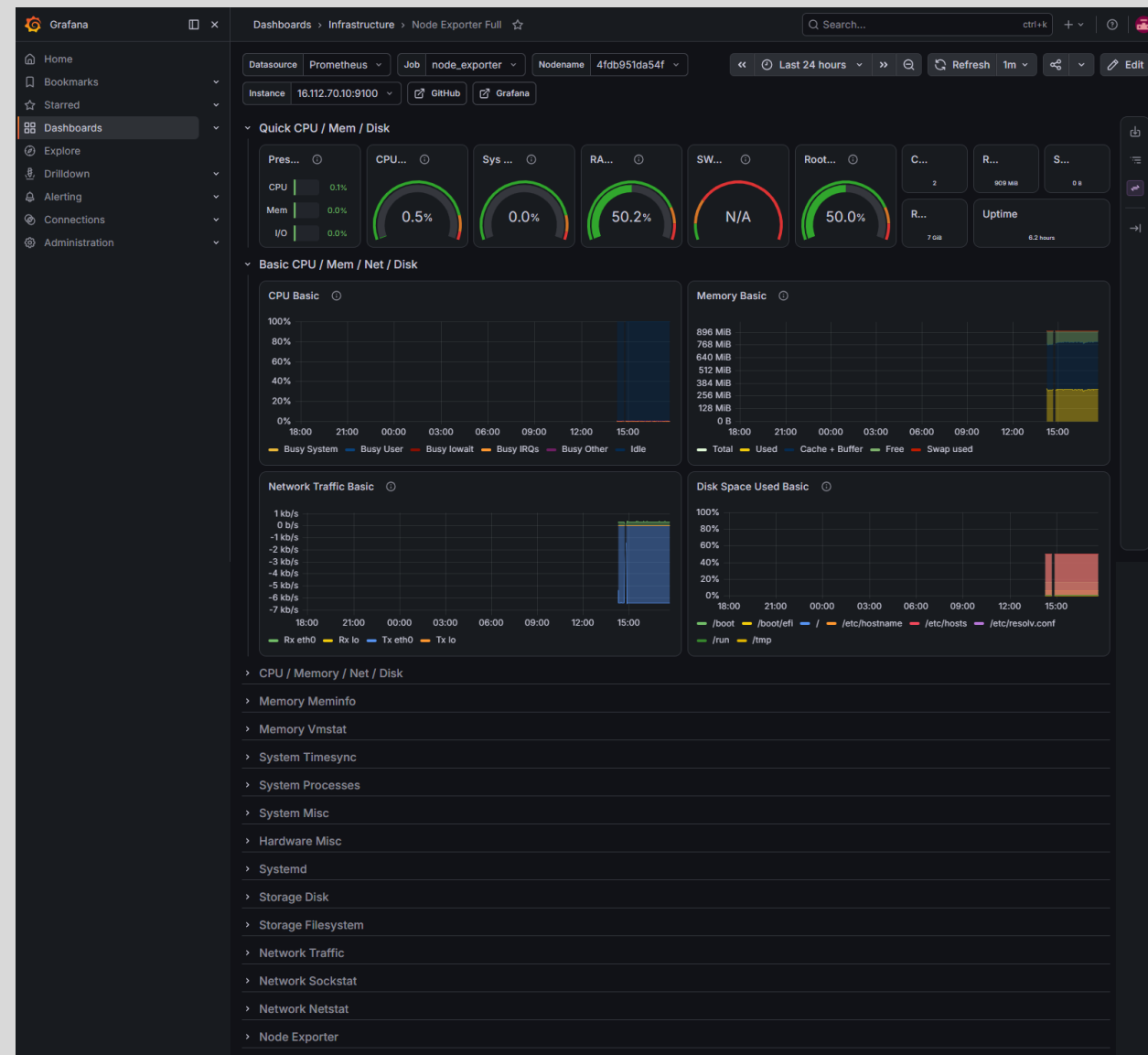
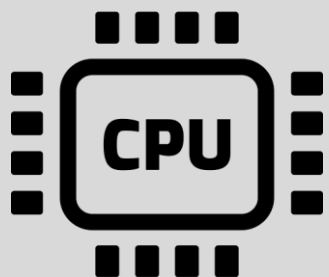
Scrape Pool	Endpoint	Labels	Last scrape	State
apache_exporter (2 / 2 up)	http://16.112.70.10:9117/metrics	instance="16.112.70.10:9117" job="apache_exporter"	10.645s ago 3ms	UP
	http://40.192.27.88:9117/metrics	instance="40.192.27.88:9117" job="apache_exporter"	8.985s ago 4ms	UP
cadvisor_monitoring (1 / 1 up)	http://cadvisor:8080/metrics	instance="cadvisor:8080" job="cadvisor_monitoring"	10.968s ago 38ms	UP
cadvisor_web (2 / 2 up)	http://16.112.70.10:8080/metrics	instance="16.112.70.10:8080" job="cadvisor_web"	12.225s ago 37ms	UP
	http://40.192.27.88:8080/metrics	instance="40.192.27.88:8080" job="cadvisor_web"	9.656s ago 35ms	UP
node_exporter (2 / 2 up)	http://16.112.70.10:9100/metrics	instance="16.112.70.10:9100" job="node_exporter"	9.749s ago 15ms	UP
	http://40.192.27.88:9100/metrics	instance="40.192.27.88:9100" job="node_exporter"	10.742s ago 15ms	UP
prometheus (1 / 1 up)	http://127.0.0.1:9090/metrics	instance="127.0.0.1:9090" job="prometheus"	10.887s ago 7ms	UP

Infrastructure Monitoring Dashboard



Key Monitoring Metrics

- ✓ CPU Utilization
- ✓ Memory Usage
- ✓ Disk Space
- ✓ Network Traffic
- ✓ Real-time Infrastructure Visualization



Alertmanager

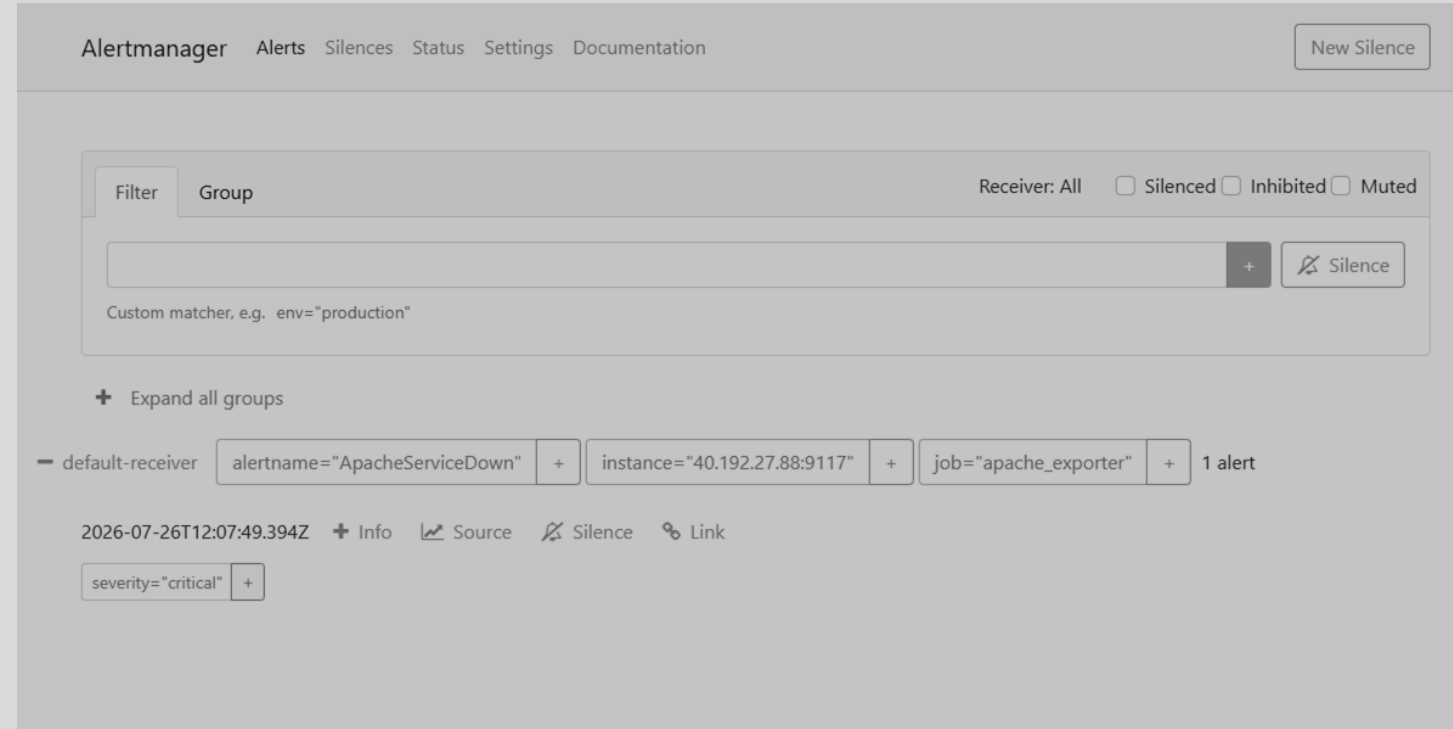


Alert Management

- ✓ Evaluates alert rules
- ✓ Detects service failures
- ✓ Sends alert notification
- ✓ Reduces downtime

Example Alerts

- ✓ Apache Service Down
- ✓ Node Exporter Down
- ✓ High CPU Usage
- ✓ Target Unreachable



Challenges Faced



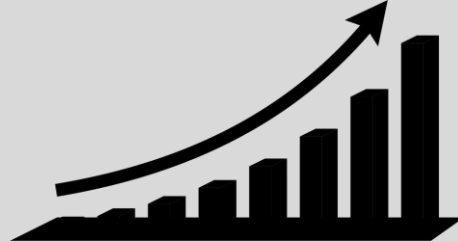
Challenge

Challenge	Solution
Dynamic Inventory Configuration	Implemented Ansible Dynamic Inventory
SSH Authentication Issues	Configured SSH Keys & Security Groups
Terraform Remote State Management	Used Terraform Cloud for Remote State
Prometheus Scrape Configuration	Validated Prometheus Targets & Metrics
GitLab CI/CD Pipeline Debugging	Optimized GitLab CI/CD Pipeline
Docker Container Networking	Configured Docker Networking & Compose



Solutions

Future Enhancements



- ✓ Kubernetes-based Deployment
- ✓ Slack & Email Alert Notifications
- ✓ Auto Scaling Support
- ✓ Centralized Log Management (Loki)
- ✓ Distributed Tracing (Tempo)
- ✓ Multi-Region Monitoring

Kubernetes → Modern container orchestration

Slack/Email → Production alerting

Loki → Centralized log aggregation

Tempo → Distributed tracing for microservices

Auto Scaling → Dynamic infrastructure support

Multi-Region → High availability and disaster recovery



Current Project

Kubernetes

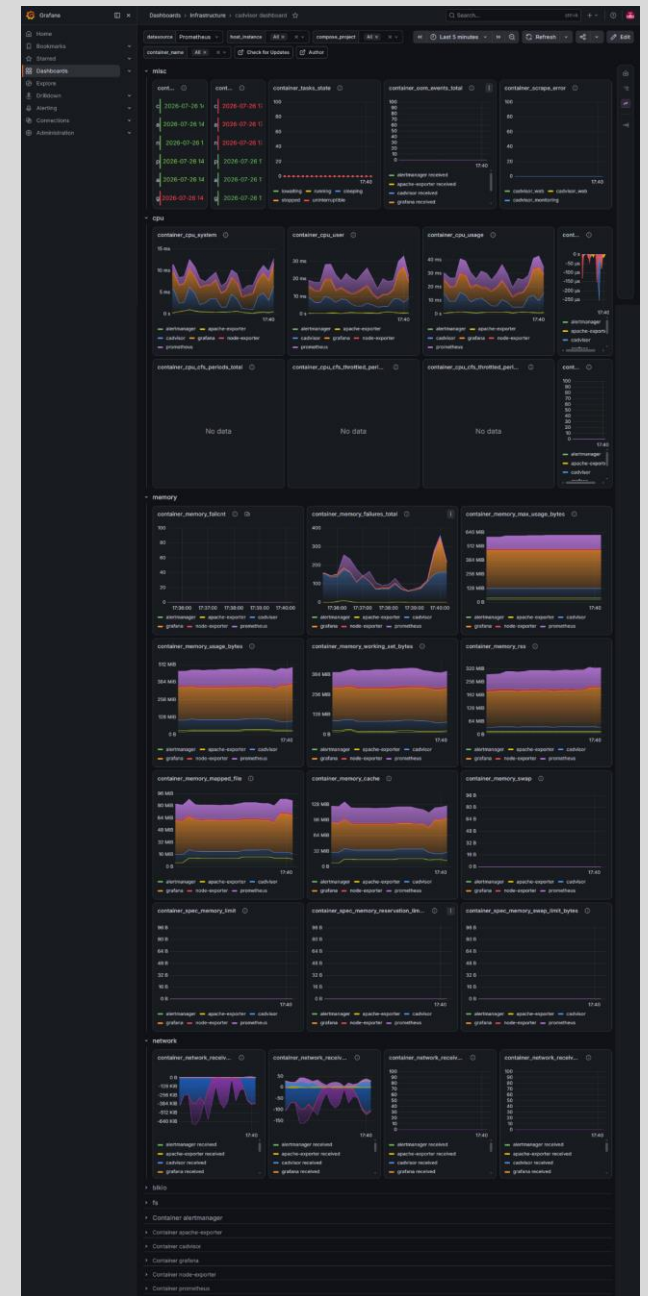
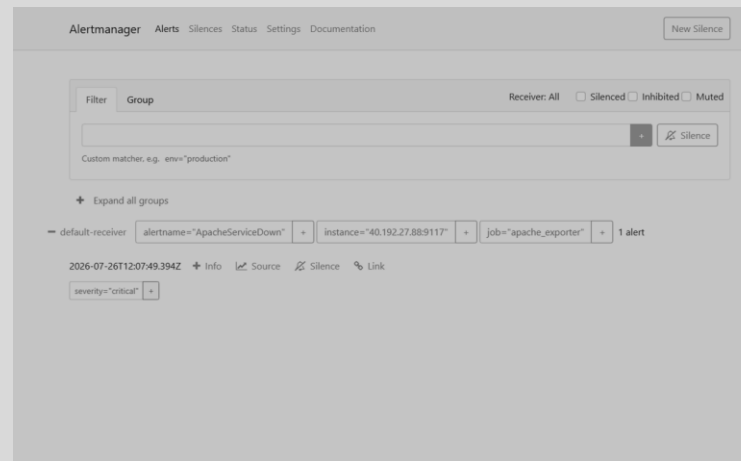
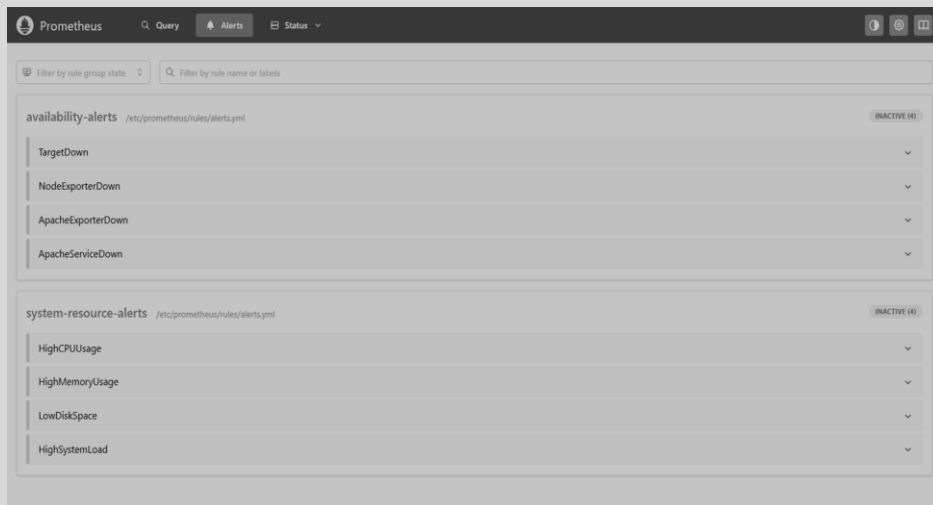
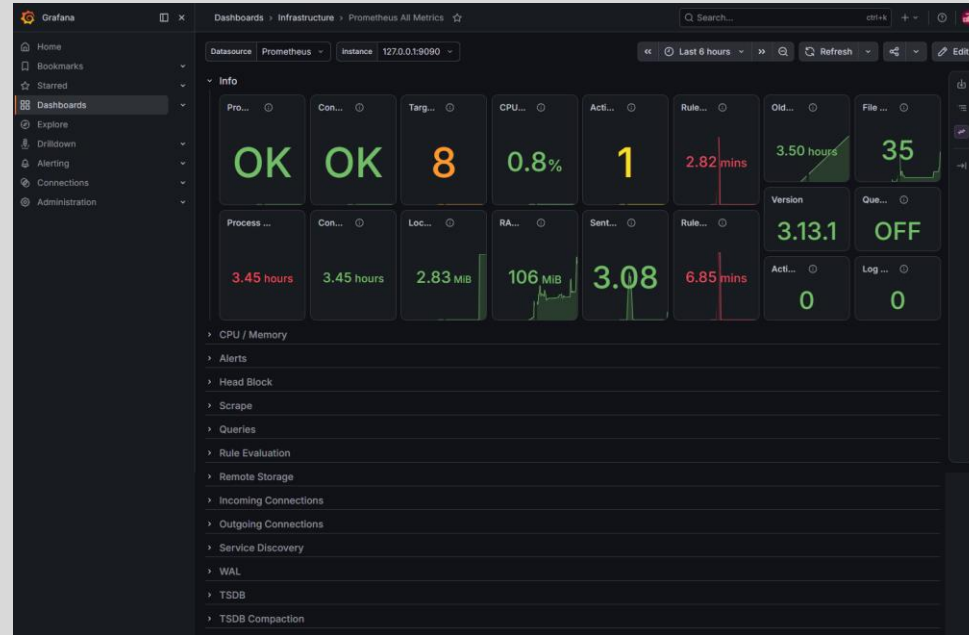
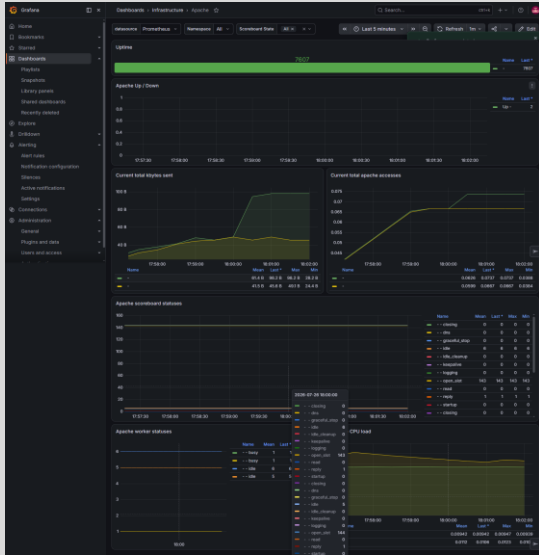
Slack Alerts

Loki + Tempo

Multi-Region
Monitoring

DEMO

Demonstration



Thank you

- Sazad Ahemad

